ADSP 31014 Statistical Models for Data Science
Course Project Part 1

Overview of Data Set:
The dataset consists of approximately 32.3k records describing employees of the City of Chicago. It was obtained from the Chicago Data Portal (https://data.cityofchicago.org/Administration-Finance/Current-Employee-Names-Salaries-and-Position-Title/xzkq-xp2w/about_data). Each observation contains information about an employee's job title, department, employment type (full-time or part-time), salary basis (annual or hourly), typical working hours, and corresponding pay rate. The data provides a detailed view of the compensation structure across different departments and positions. Salaried and hourly employees are both represented, with pay information reflecting either annual or hourly compensation.

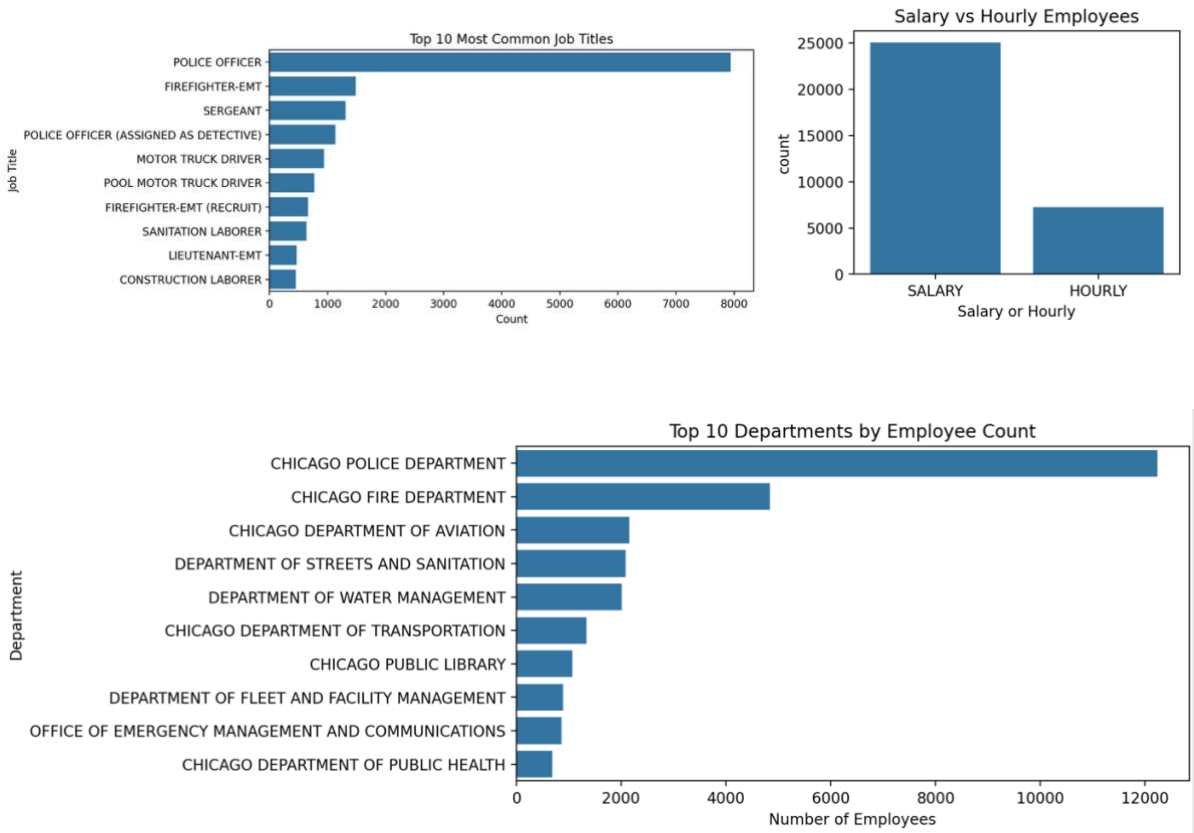

Figure 1: Top 10 most common job titles (upper left). Top 10 departments by employee count (low). Proportion of salaried and hourly employees (upper right).

Data Table Schema:
The data set consists of 32,278 rows and 8 columns. Each row represents one record of a City of Chicago employee.

Column Name	Description	Data Type	Example Value	Notes
Name	Name of employee	string	“ABAD JR, VICENTE M”	● Not unique, some employees have the same names.
Job Titles	Title of employee at the time when the data were updated	string	“CIVIL ENGINEER IV”	● 1199 unique job titles.
Department	Department where employee worked	string	“DEPARTMENT OF WATER MANAGEMENT”	● 39 unique departments.
Full or Part-Time	Whether the employee was employed full-(F) or part-time(P)	string	“F”	● 3 rows of missing value.
Salary or Hourly	Defines whether an employee is paid on an hourly basis or salary basis	string	“SALARY”	● 77.5% salaries, 22.5% hourly.
Typical Hours	Describes the typical amount of work for hourly employees. The data does not apply to salary employees.	decimal	40.0	● 25,020 rows of missing data for employees paid by annual salary. ● Most hourly employees work 40 hours per week.
Annual Salary	Annual salary rates. Only applies for employees whose pay frequency is “Salary”.	decimal	$137,436.00	● 7258 rows of missing data for hourly employees.
Hourly Rate	The hourly salary rates for individuals whose pay frequency is “hourly”.	decimal	$51.40	● 25,020 rows of missing data for employees with annual salary. ● Individuals participating volunteer programs get a very low hourly rate ($4.73).

Data Cleaning and Processing:
The two main data quality issues are inconsistent data types (caused by dollar signs) and duplicated employee records. Missing values follow a clear pattern: Annual Salary is missing for hourly employees, while Hourly Rate is missing for salaried employees. These missing values are by design rather than random since each pay variable applies only to one type of employee.

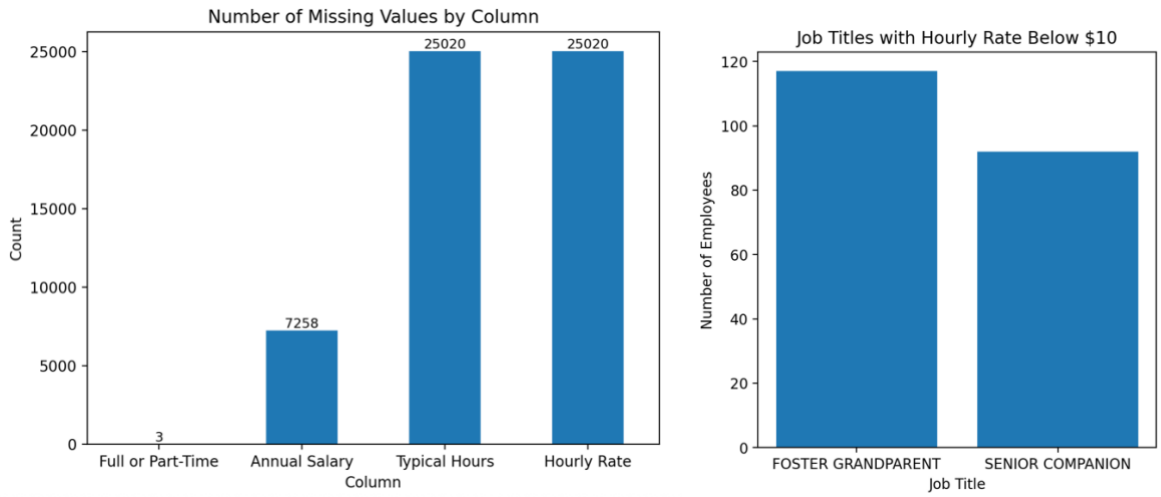

Figure 2: Number for missing values by column (left). Job titles with hourly rate below $10 (right).

Consequently, the data set is cleaned through following steps:

- Removed dollar symbols and standardized numeric columns (“Annual Salary” and “Hourly Rate”).
- Removed fully duplicated rows.
- Dropped rows with missing “Full or Part-Time” type.
- Removed records of individuals participating in volunteer programs, whose hourly rate $4.73 is unrealistically low. These volunteers are not formal employees at City of Chicago and were excluded to avoid distorting later analysis of salary-related factors.

The resulting cleaned data set has 32,055 rows.